



## I2C interface for LCD

Product Code: LCD2004

Availability: In Stock

**R95.00**

Qty

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Description

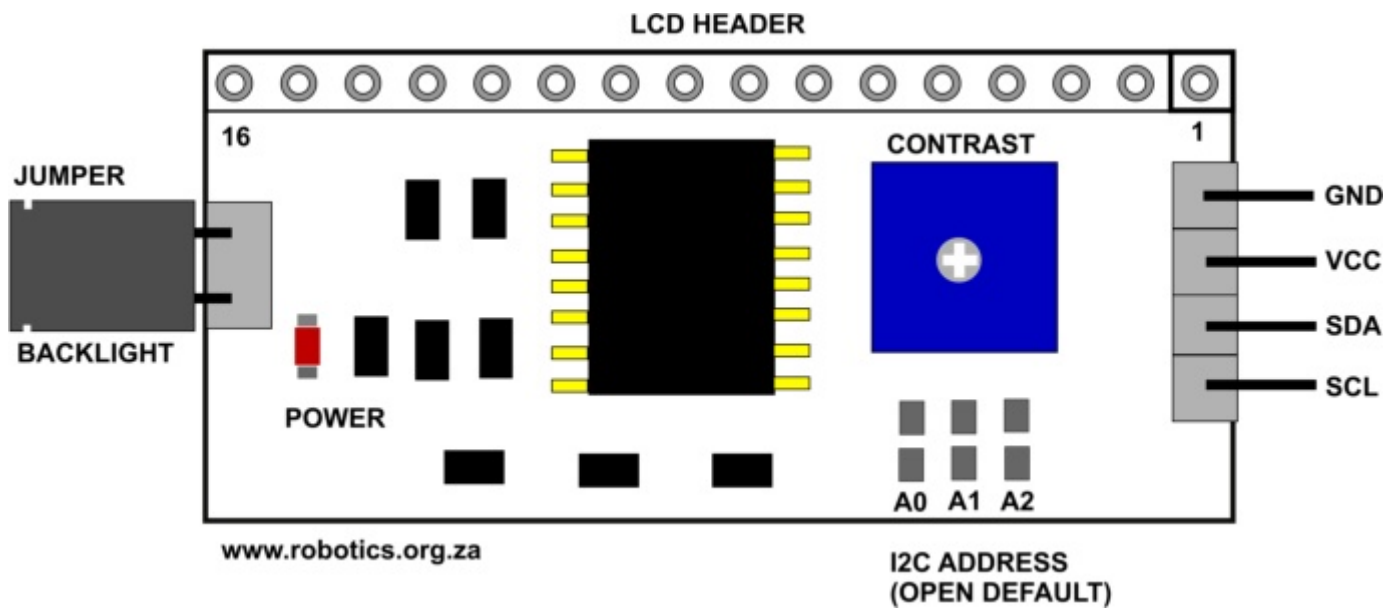
### I2C interface for LCD

This LCD2004 is a great I2C interface for 2x16 and 4x20 LCD displays. With the limited pin resources, your project may be out of resources using normal LCD shield. With this I2C interface LCD module, you only need 2 lines (I2C) to display the information. If you already has I2C devices in your project, this LCD module actually cost no more resources at all. Fantastic for Arduino based projects.

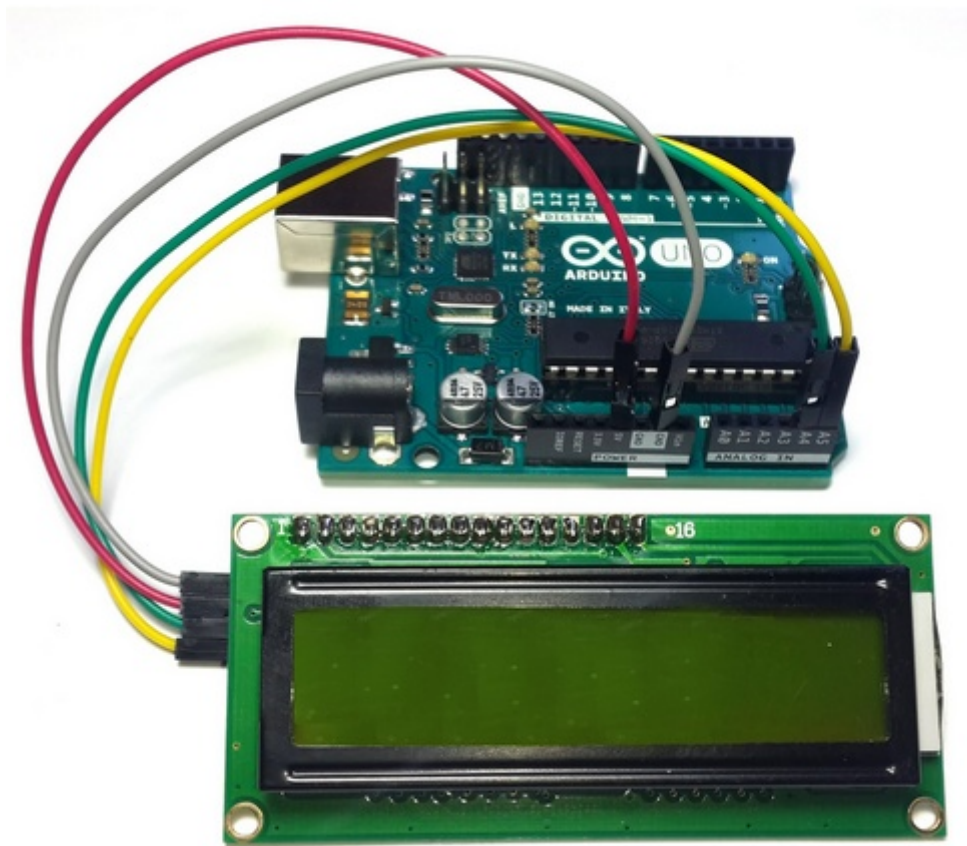
### Specification

- ▶ Compatible with 16x2 and 20x4 LCD's
- ▶ Default I2C Address = 0X27
- ▶ Address selectable - Range 0x20 to 0x27

### Board Layout



### I2C Address Setup

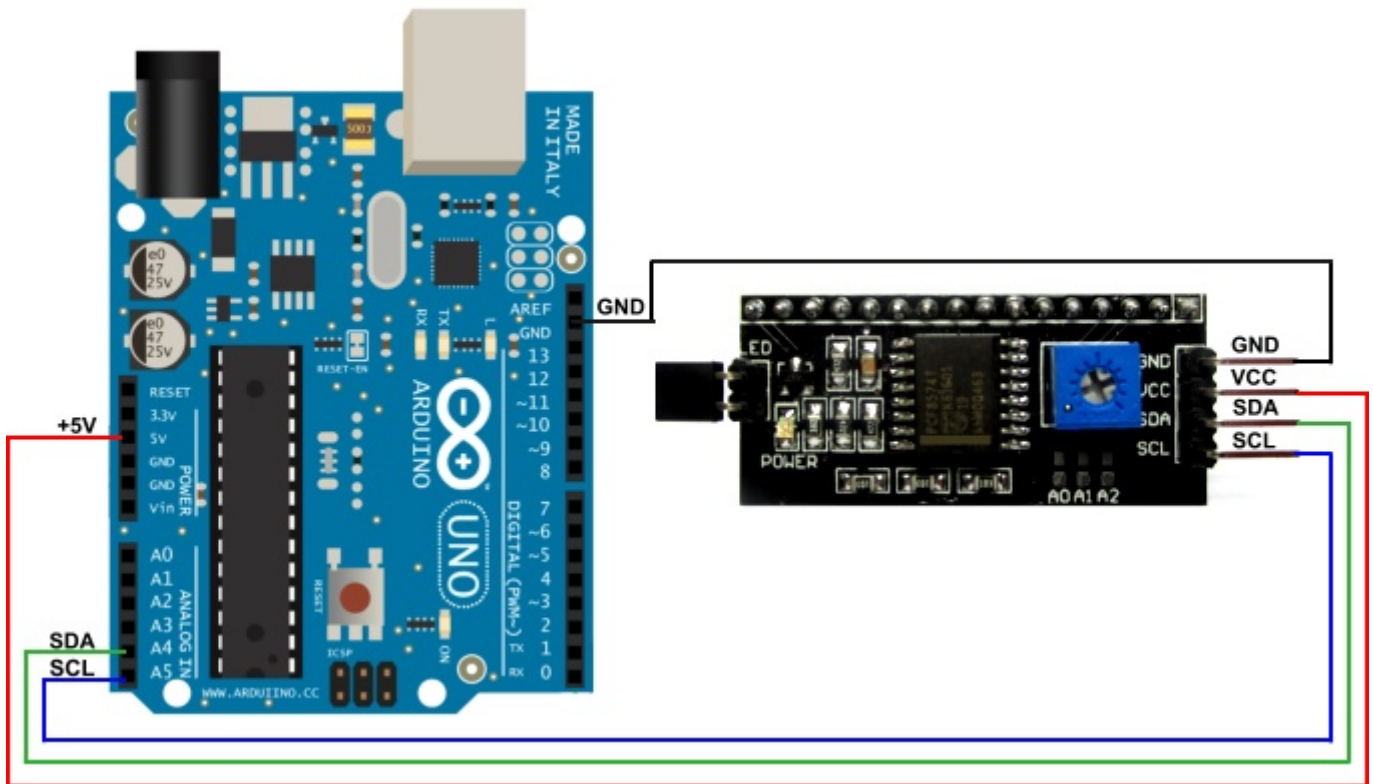
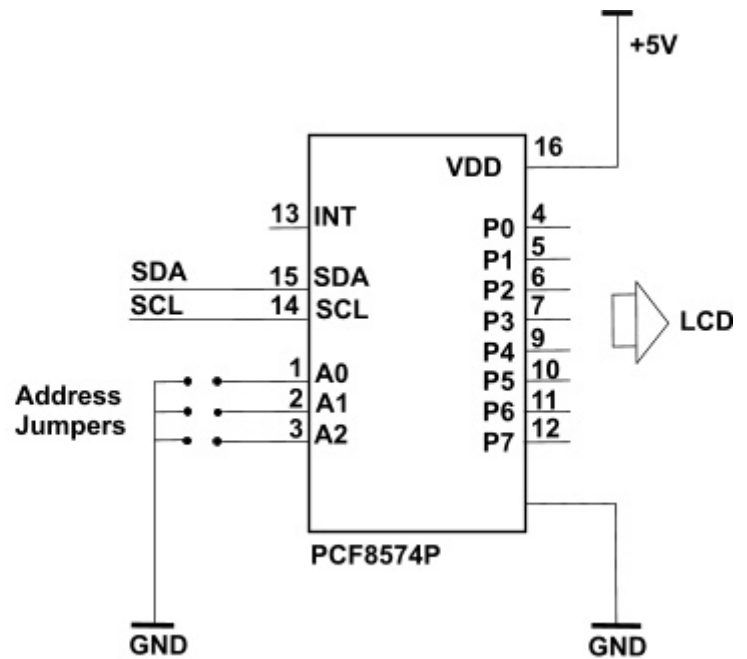


The LCD2004 board utilized the PCF8574 I/O expander. This nifty little chip provides eight bits of parallel I/O addressable by a I2C bus address – 0x00 to 0x27. SainSmart tied all address leads to Vcc, so the LCD2004 board's I2C address is permanently fixed at hex 27. This is rather limiting since no additional LCD2004s can be added to the bus. Anyway, you simply address the board and write an eight bit value which is then presented on the output pins of the PCF8574, which, in this case, are connected to the HD44780 based LCD screen.

| INPUTS |    |    | I2C SLAVE ADDRESS |
|--------|----|----|-------------------|
| A2     | A1 | A0 |                   |
| L      | L  | L  | 0x20              |
| L      | L  | H  | 0x21              |
| L      | H  | L  | 0x22              |
| L      | H  | H  | 0x23              |
| H      | L  | L  | 0x24              |
| H      | L  | H  | 0x25              |
| H      | H  | L  | 0x26              |
| H      | H  | H  | 0x27              |

H = Open Jumper    L = Close Jumper

### Controlling your I2C Display with Arduino



### Arduino Code

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
// Set the LCD address to 0x27 for a 16 chars and 2 line display
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
void setup()
```

```
{  
  
  // initialize the LCD  
  lcd.begin();  
  
  // Turn on the backlight and print a message.  
  lcd.backlight();  
  lcd.print("Hello, world!");  
}  
  
void loop()  
{  
  // Do nothing here...  
}
```